Management of severe upper GI bleeding

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Presentation of upper GI bleed

- Haematemesis
- Coffee ground vomiting (methaemoglobin)
- Melaena (haematin) (also named tarry stool)
- Fresh PR bleed (also named haematochezia)
- Occult bleed (symptoms of anaemia)

Spelling

- Melaena (UK)
- Melena (US)

M~~a~~laena

Acid $\rightarrow$ Hb $\rightarrow$ Methaemoglobin (Coffee ground)

Gut bacteria $\rightarrow$ Hb $\rightarrow$ Haematin (Melaena)

Melaena – fresh vs old

Fresh melaena
= Hb + haematin

Old melaena
= haematin

Severity according to presentation

Ongoing / severe	Slow / bleeding stopped
Haematemesis	Coffee ground vomiting
Fresh melaena	Old melaena

Causes of upper GI bleeding

In descending order of frequency:

- Duodenal or gastric ulcer
- Gastritis
- Esophageal or gastric varices
- Mallory-Weiss syndrome
- Benign or malignant gastric tumour

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✓ = more likely severe

Causes of upper GI bleeding

- Esophagitis, esophageal tumour
- Stomal ulcer
- Aortoduodenal fistula
- Haemobilia, haemosuccus pancreaticus
- Vascular malformation, angiodysplasia
- Dieulafoy's lesion
- Duodenal or jejunal diverticulum, jejunal ulcer

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History

- Nature, rate, and duration of bleeding
- Previous episode(s)
- Pain
- Weight loss, anorexia
- Medical conditions, liver disease, HBsAg, bleeding tendency
- History of irradiation (enteritis)

History

- Alcohol
- Drug history
  - Aspirin, NSAIDs
  - Anticoagulant (e.g. warfarin)
  - Antiplatelet agent (e.g. aspirin, clopidogrel)
  - Cardiac drugs – beta blockers
  - Iron – black stool

Signs

- Pallor, tachycardia, hypotension
- Cervical lymph node
- Signs of chronic liver disease
- Caput medusae, splenomegaly, shifting dullness
- Abdominal mass
- PR exam – fresh and old melaena

Management

- Resuscitation - early recognition of shock, earliest sign is tachycardia
- Diagnosis - history, examination, investigation
- Treatment

Blood loss and shock

	Class I	Class II	Class III	Class IV
Blood loss (%/ml)	< 15%/ 750	15 – 30%/ 750 – 1500	31 – 40%/ 1501 – 2000	> 40%/ 2000
Heart rate (bpm)	↔	↔ / ↑	↑	↑ / ↑↑
Blood pressure	↔	↔	↔ / ↓	↓
Pulse pressure (SBP – DBP)	↔	↓	↓	↓
Respiratory rate	↔	↔	↔ / ↑	↑
Urine output (ml/h)	↔	↔	↓	↓↓
GCS	↔	↔	↓	↓
Base deficit (mmol/L)	0 to -2	-2 to -6	-6 to -10	<= -10
Need for transfusion	Monitor	Possible	Yes	Massive

Resuscitation

- Secure ABC
- Nil by mouth, NG tube in selected patients
- Large bore IV cannula
- Group O Rh-ve blood, colloids / crystalloids
- Type & x-match, haemocue, CBP, R/LFT, PT/APTT
- Erect CXR

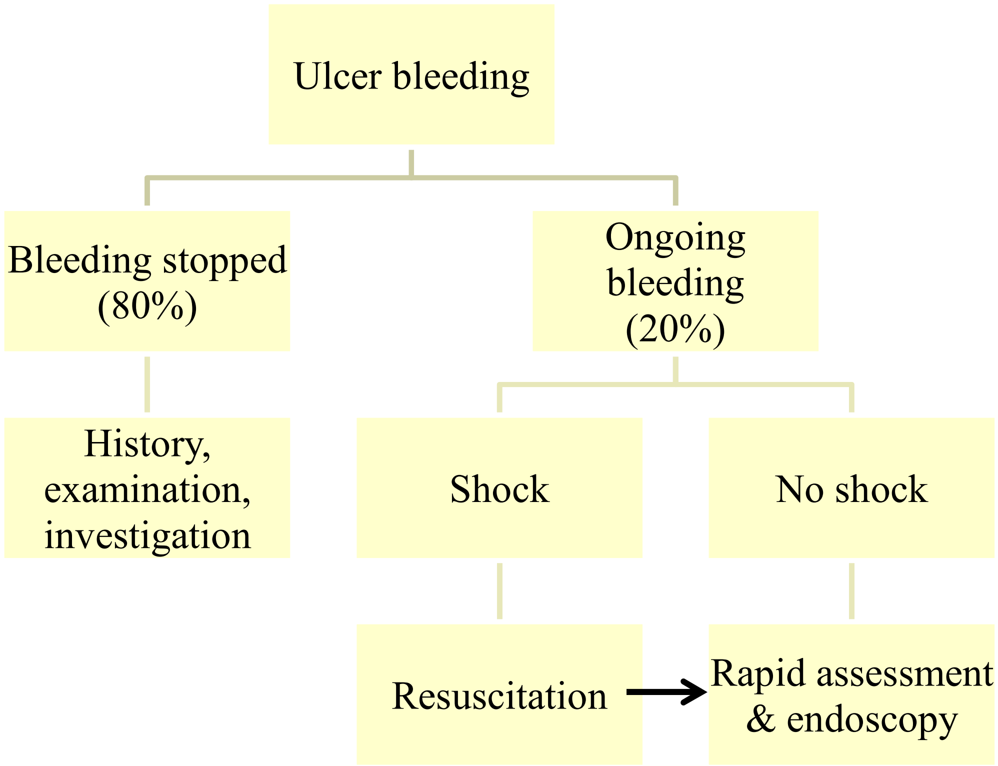
Monitoring

- Shock chart hourly
- Blood pressure and pulse
- Respiratory rate, pulse oximeter (SaO_2)
- Core temperature
- Urine output ($\geq$ 0.5 ml/Kg/hr)
- Central venous pressure (CVP)
- Cardiac monitor

Diagnosis

- Ulcer bleed stops spontaneously in approx 70-80%
- Identify pt in shock
- Identify pt having ongoing bleeding

General guideline



Features suggestive of ongoing bleeding

- Haematemesis
- Fresh melaena
- Tachycardia
- Fresh per rectal bleeding
- Fresh blood aspirated from NG tube

Role of upper endoscopy

- Verify bleeding source
- Stratify risk of rebleeding
- Therapy - definitive, temporizing

Endoscopic stigmata

ESRH (Forrest's classification)

Ia	spurting	80-90%
Ib	oozing	30%
IIa	nonbleeding vv	20-50%
IIb	adherent clot	20-30%
III	clean base	0-2%

Acute treatment

- Depends on diagnosis
- Bleeding peptic ulcer

Bleeding peptic ulcer

- Clean base – start feeding, early discharge
- Therapeutic endoscopy
  - Injection method: adrenaline
  - Thermal method: heater probe
  - Mechanical method: metal clip
- H2 blocker, PPI
  - hasten healing of ulcers
  - PPI infusion

After therapeutic endoscopy

- Close monitoring
- Look out for rebleeding (first 3 days)
- Signs of possible rebleeding
  - Increasing pulse rate
  - Haematemesis
  - Pass fresh melaena again
  - Fresh blood aspirated from nasogastric tube
  - Drop in haemoglobin level

Risk factors for recurrent bleeding

- Shock on presentation
- Hb < 8.0 g/dl on presentation
- Transfusion
- Age > 60 yrs
- Comorbidity
- Coagulopathy

Risk factors for recurrent bleeding

- Patient already hospitalised
- Large ulcer
- Ulcer on posterior D1
- Ulcer on higher posterior lesser curve

Surgery for bleeding ulcer

- Indications
  - therapeutic endoscopist not available
  - massive bleeding
  - failed endoscopic therapy
  - rebleed after endoscopic therapy
- Plication of bleeder + additional procedure

Choice of additional procedure

- Condition of patient
- Experience of surgeon
- Type of ulcer
  - DU: V+P
  - GU: partial gastrectomy